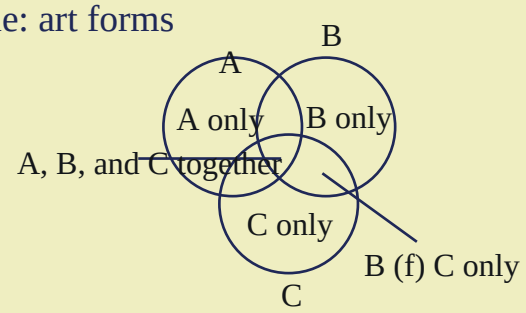
1 Art forms

Rule: art forms

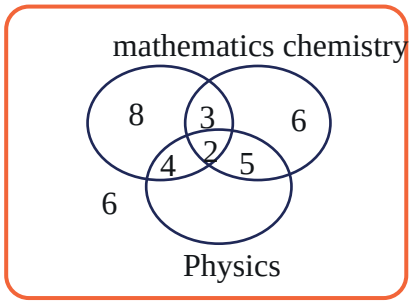


(or) means union, (f) means intersection

1 The following figure shows the number of participants in a competition Mathematics, Chemistry and Physics Answer the questions

What's next? A:

Number of subscribers to all three articles together ?



A 5 B 4 C 2 D 1

Answer: C

Existing number In the common part between the three departments = 2

(Number of participants in mathematics only? 1) in the question

A 4 B 6 C 5 D 8

Answer: D

Existing number In the Mathematics Circle section, there is no

8 = other circles

(Number of participants in mathematics 1) in the question And chemistry?

A 4 B 5 C 6 D 9

Answer: B

Existing number In the joint part of the Mathematics Department

5 = And the chemistry department

(Number of participants in mathematics 1) in the question And just chemistry?

A 6 B 5 C 4 D 3

Answer: D

Existing number In the joint part of the Mathematics Depart

3 = Chemistry department only and does not include physics

(Number of participants in mathematics 1) in the question Or chemistry?

A 18 B 19 C 28 D 21

Answer: C

Existing number In the Department of Mathematics and the

= 28

(Number of participants in the three articles? 1) in the quest

A 15 B 18 C 20 D 28

Answer: D

Existing number In the Department of Mathematics and the

The Department of Chemistry and the Department of Physic

(1) How many students are not participating in the question Third articles?

A 8 B 7 C 6 D 4

Answer: C

6 = number of students outside the circles

(1) How many students participated in the question? Did you only need me?

A 8 B 12 C 14 D 15

Answer: B

Common 2 Students at the intersections of circles and not ac

12 = And the survivor is

(How many students are there? 1) in the question

A 14 B 24 C 34 D 50

Answer: C

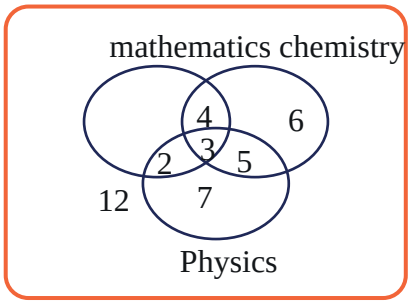
34 = the sum of all students on the drawing

Legend training

The following figure shows the number of participants in a competition Mathematics, Chemistry and Physics Answer the questions

What's next? A:

Number of subscribers to all three articles together ? !



(A) 3 (B) 2 (C) 1 (D) zero

In the training question, the number of participants in mathematics Only?

(A) 3 (B) 2 (C) 1 (D) zero

In the training question, the number of participants in mathematics And chemistry?

(A) 2 (B) 3 (C) 7 (D) 20

In the training question, the number of participants in mathematics And chemistry only?

(A) 7 (B) 6 (C) 5 (D) 4

In the training question, the number of participants in mathematics Or chemistry?

(A) 22 (B) 21 (C) 20 (D) 18

In the training question, the number of students participating in Third articles?

(A) 27 (B) 18 (C) 16 (D) 14

In the training question, how many students are there? Participants in the three subjects?

(A) 10 (B) 12 (C) 13 (D) 14

In the training question: How many students participated in Did you only need me?

(A) 9 (B) 10 (C) 11 (D) 14

In the training question: How many students?

(A) 32 (B) 35 (C) 39 (D) 40

(2) 240 students and participants in the school subject

In subjects 8 in Physics and 20 in Sixth Mathematics

Other How many are not participating in any subject?

(A) 140 (B) 160 (C) 172 (D) 90

Answer: C

6 = one-sixth of the student 1 x 240 = 40 Math

Students participating in the subjects = 68=8+20+40

∴Number of non-participants ··· = ·· - ··· =

Legend training

Students and participants in 240 schools in it

In the subjects 28 in Physics and 40 in Mathematics, the qua

Other How many are not participating in any subject?

(A) 112 (B) 160 (C) 382 (D) 292

(3) A student divided the student teacher into 25 classes

Student, how many 18 students and an oath by 12, an oath b

The number of students in both sections together ?

(A) 3 (B) 4 (C) 5 (D) 6

Answer: C

plural = 12 + 18 = 30

· = ·· - ·· =The participants are the difference

Legend training

Student The teacher divided the student into two sections with 20 classes

Student, how many students and a section with 10 sections?

Students who are in both sections together ?

(A) 5 (B) 4 (C) 3 (D) 2

(4) Call him Of them, there were 30 participants in the ball, and 50 participants

Not participating: 7 participants in swimming and 26 foot participants

In either game. How many subscribers in the past?

And tourism?

(A) 0 (B) 13 (C) 14 (D) 15

Answer: B

others the subtract We = 50 - 7 = 43

participants of pool = 30 + 26 = 56

games two the in participants = 56 - 43 = 13

Legend training

Of them, 40 participated in football, and 60 participated in its club

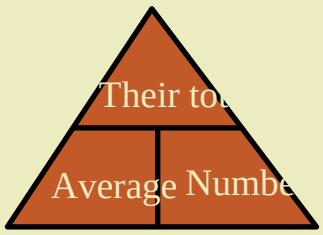
Not participating in any of the 8 participants in swimming and 26 and

In the two games, how many participants participate in football and swimming?

(A) 0 (B) 14 (C) 15 (D) 16

2 Measures of central ten

(1) Arithmetic average Average - Mean -



(A) Average = $\frac{\text{Their total}}{\text{Their number}}$

Their sum = number x average

(5) What is 183 = the sum of three integers

Average?

(A) 61 (B) 62 (C) 63 (D) 64

Answer: A

total = 183 3,

= average $\frac{\text{Total pi} \quad \text{Annat}}{\text{Their number}} = \frac{183}{3} = 61$

Legend training

What is 160 = the sum of four integers

Average?

(A) 30 (B) 40 (C) 45 (D) 50

(6) E If the average numbers , S, pp. 25, 11

Y?+ Then find the value of x18 equals

(A) 36 (B) 48 (C) 50 (D) 72

Answer: A

Calculate the total numbers

Total = average x number = 18 x 4 = 72

∴S Y+ = 72 - (25+11) = 36

Legend training

, x, p. 27, 12 if the average of the numbers

y?+ x+ 27 + 12, so find the value of 12 equals

A 24 B 36 C 38 D 48

7 If the elevator is 900 kg, it can carry

The arithmetic average of the weights of the company·s employees

kg. How many people can it carry? 75 is

A 10 B 12 C 15 D 16

Answer: B

total 900, = average 75, =

Their number = $\frac{\text{Total pi}}{\text{Average}}$ Annat = $\frac{900}{75}$ 12 =

Legend training

kg. If the average is 1200, an elevator can carry

60 The calculation of the weights of the company·s employees is

kg, how many people can it carry?

A 10 B 20 C 22 D 25

8 You died Four whose first term is equal to Alone the second 1

Starting from the third term, every term in it equals 5

What is the arithmetic mean of all previous terms?

Is it twenty-five?

A 2.5 B 3 C 5 D 25

Answer: B

I was bored Four =, ..., limit the third $\frac{5+1}{2}$ 5, 1 = 3

=, define the fourth $\frac{3+5+1}{3}$ = 3

3 = And so the situation continues throughout the borders

Legend training

You died Four whose first term is equal to Only t

7 equals Starting from the third term, every term i

The arithmetic mean of all previous terms, and w

Twenty-fifth?

A 2.5 B 4 C 5 D 25

Average for x, y compare brown1 + if m

The first value: $\frac{+p \quad S}{2}$ second value. A: M

A The first value is greater B The second value I

C The two values ·· are equal D Data are insuff

Answer: A

Average for x, y1 + m

$\frac{+p \quad S}{2}$ m = > M1+

The first value is greater

Legend training

If m is the average of x, y compare brown

The first value: $\frac{+p \quad S}{2}$ second value. A: M

A The first value is greater B The second value I

C The two values ·· are equal D Data are insuff

Other numbers 4 but 8 numbers among them is the Ha

Calculate the average for all 3? The arithmetic mean

A 4 B 5 C 6 D 7

Answer: C

48 = 8 6 = 6 numbers, the sum of the

of sum numbers, 4 = 4 3 = 12

Average of all = $\frac{12+48}{10} = \frac{60}{10} = 6$

Legend training

Other numbers are 4, but 12 are numbers, the middle of which is the arithmetic mean

Calculate the average for all 7? The arithmetic mean

A 4 B 5 C 10 D 11

B New average = old average + $\frac{\text{Increase}}{\text{Their number}}$

11 If the teacher finds out that 88 students have 10 average grades

A grade, so he added 20 to him. He made a mistake in the grades and found that a student had made a mistake

Compare between:

The first value: the average after adjustment.

The second value Oh 91 :

A The first value is greater B The second value It's bigger

C The two values ∵are equal D Data are insufficient

Answer: B

average old = 88 = increase 20, =

The new average = ∵ + $\frac{20}{10}$ = 90

∴The second value It's bigger

Legend training

Average grades If the teacher discovers an error, 40 = 10 students

A grade, so he added 10 to his grades and found that a student was his

Compare brown

The first value is the average after adjustment,

The second value Oh ∵ :

A The first value is greater B The second value It's bigger

C The two values ∵are equal D Data are insufficient

12 50 = If the average weights of a number of children

kg became 62 kg, and when adding Saad, who weighed

kg, how many 52 = average weights of children

Children?

A 4 B 5 C 6 D 7

Answer: C

52 = new average, 50 = old average

luck Increase = 62 - 50 = 12

+∵ = ∵ $\frac{12}{S}$ Solve the equation

$\frac{2}{1} = \frac{12}{S}$ ⟶ x2 = 12

6=S

Legend training

K ff 45 = if the average weights of a number of children

The kg became 65 and when adding Muhammad·s weight

kg, how many children are there? 50 = average

A 4 B 5 C 6 D 7

C Numbers with equal increases or decreases

= average $\frac{\text{+ The first The other}}{2}$ = limit the middle

13 Compare brown

Mechanism 5 First value: average Matthew num

The second value A: The third issue

A The first value is greater B The second value

C The two values ∵are equal D Data are insu

Answer: C

Average of five numbers It is the middle numbe

= the third

The two values ∵are equal

Legend training

Compare brown

Message 7 The first value is average Matthew numbers

The second value A: The third issue

(A) The first value is greater (B) The second value It's bigger

(C) The two values ·· are equal (D) Data are insufficient

---

(14) Find the arithmetic mean of the numbers
1350 , 1375 , 1400 , 1425 , 1450 , 1475

(A) 1412.5 (B) 1415 (C) 1425 (D) 1430

Answer: A

The numbers are amazing Cut Constantly25

$$\text{Average} = \frac{1350 + 1475}{2} = \frac{2825}{2} = 1412.5$$

Legend training

Find the arithmetic mean of the numbers
1370 , 1390 , 1410 , 1430 , 1450 , 1470

(A) 1420 (B) 1422 (C) 1425 (D) 1430

---

(15) The mechanism in their midst5 Odd numbers are dead What is the middle 15

The first three numbers?

(A) 11 (B) 12 (C) 13 (D) 15

Answer: C

The numbers are dead Mechanism
The average is the middle limit
, ... , ... 15 , ... , ...
19, 17, 15, 13, 11 numbers
·· = Average of the first three

Legend training

Their midst 5 Odd numbers are dead What is the middle to t
Three numbers?

(A) 11 (B) 12 (C) 15 (D) 19

---

(16) The mechanism of their sum 5 Even numbers are met S

The youngest?

(A) 50 (B) 56 (C) 60 (D) 62

Answer: B

total = 300 = 5

$$= \text{Average} \quad \frac{300}{5} \cdot\cdot =$$

The numbers are 64 , 62 ,60 , ·· , ·· :

∴Smaller number ·· =

Legend training

The mechanism of their sum 5 Even numbers are met What

(A) 44 (B) 40 (C) 36 (D) 22

(D) The average between two even numbers is an odd nu

(A) The average between two odd numbers is an even

(17) If the average of four even numbers

You died Why is n, the largest of these numbers is?

(A) 2-n (B) 3-n (C) 4-n (D) 3 +n

Answer: D

The numbers are even Mechanism
Their average is the middle number and is an odd number
n is an odd number
3 +, n1+, n, n1 -, n3 - The numbers are: n
The largest number is n · +
You can experiment with values

Legend training

If the average of four odd numbers is... What is it?

1+ nThe largest of these numbers is?

(A) 2-n (B) 3-n (C) 4-n (D) 4 +n

(18) Company with it SAR 2000 employees, each with a salary of 10

4,000 employees, each with a salary of 3 Layer and tidy manager

Riyals, what is the average salary of employees? 10,000

(A) 2000 (B) 2500 (C) 3000 (D) 3500

Answer: C

We find the total salary at

total = 10 x 2,000 + 3 x 4,000 + 1 x 10,000

= 20000 + 12000 + 10000

= 42000

= average $\frac{42000}{14}$ = 3000

Legend training

3 riyals and 200 paid employees, each of whom has 12 companies

300 employees, each with a salary Layer and tidy manager

SAR, what is the average salary of employees? 3100

(A) 200 (B) 250 (C) 300 (D) 400

(19) Compare the numbers from 51 to 58

First value: average of even numbers

The second value A: The average of individual numbers

(A) The first value is greater (B) The second value It's bigger

(C) The two values are equal (D) Data are insufficient

Answer: A

= Individual average $\frac{57 + 51}{2} = \frac{108}{2} = 54$

= Marital average $\frac{58 + 52}{2} = \frac{110}{2} = 55$

The first value is greater

Legend training

Compare the numbers from 49 to 42

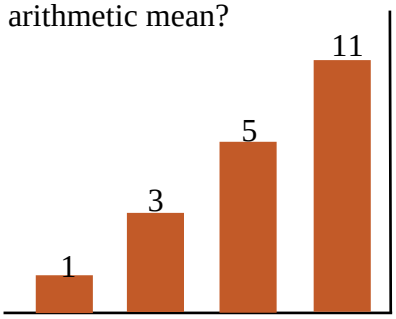
The first value is the average of even numbers

The second value A: The average of individual numbers

(A) The first value is greater (B) The second value I

(C) The two values are equal (D) Data are insuf

(20) Find the arithmetic mean?



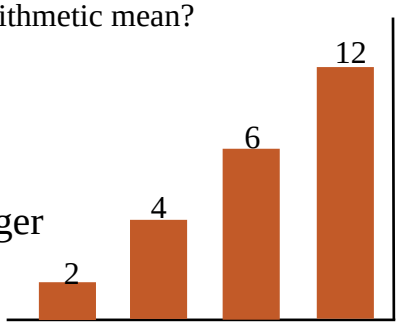
(A) 3 (B) 4 (C) 5 (D) 6

Answer: C

= average $\frac{11 + 5 + 3 + 1}{4} = \frac{20}{4} = 5$

Legend training

Find the arithmetic mean?



(A) 3 (B) 4 (C) 5 (D) 6

21 What is the degree in biology?

The material	100 degree of
90Mathematics	
95Physics	
90Chemistry	
alive	?
92 average score	

A 90 B 91 C 92 D 93

Answer: D

, total = 4? , number 92 = average

Number x Average = Total = 92 x 4 = 368

Biology = 368 - 90) + 95 + (90 = =275-368 93

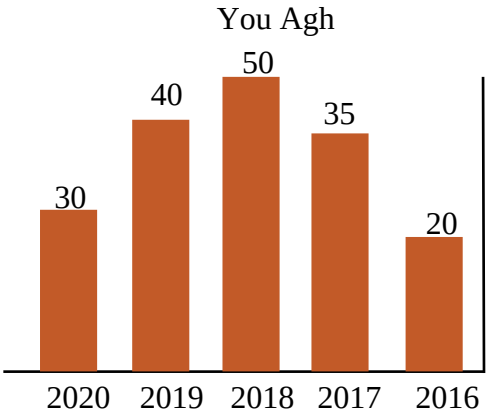
Legend training

What is the degree in biology?

The material	100 degree of
mathematics	70
Physics	65
chemistry	40
alive	?
Average grades	60

A 60 B 62 C 65 D 75

22 What another average Years?3



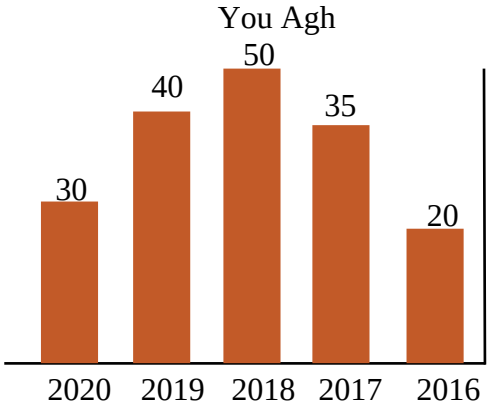
A 30 B 40 C 50 D 60

Answer: B

$$\frac{50 + 40 + 30}{3} = \frac{120}{3} = 40$$

Legend training

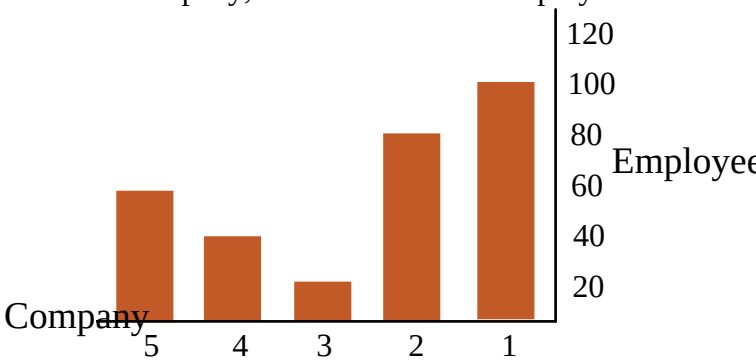
What an average first Years?3



A 30 B 35 C 52.5 D 60.5

23 200,000 riyals to my employees if a bonus is distribute

The fourth company, how much is each employee·s share?



A 4000 B 5000 C 6000 D 6500

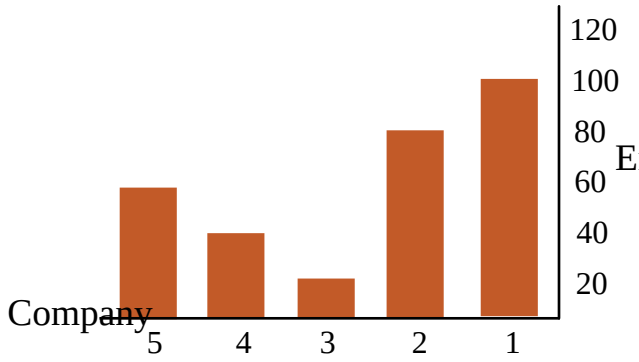
Answer: B

$$= \text{average } \frac{200000}{40} = 5000$$

Legend training

200,000 riyals to the company·s employees if a bonus is distributed

The second How much is each employee·s share?



(A) 2000 (B) 2500 (C) 6000 (D) 6500

(2) Amladi
Largest value = range - Minimum value

(2) Range of ages How old are 25 people? 5

The first and the last?

(A) ··,·· (B) 42,25 (C) 64,22 (D) 60,35

Answer: D

We search the options for the two numbers

25 The difference between them

35 and 60 are both numbers

Legend training

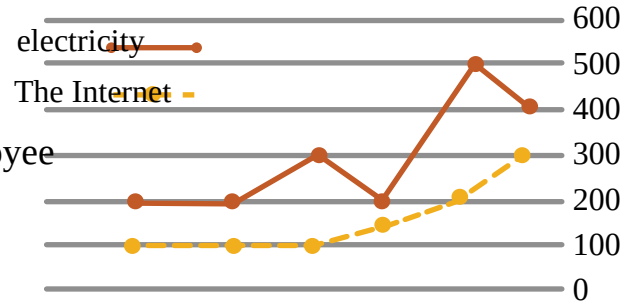
The age range of a number of people is How much is 30?

The age of the first and the last?

(A) ··,·· (B) 55,25 (C) 64,22 (D) 65,30

(3) Loom
The mode is the most frequent value

(25) What is the pattern of Internet bill values?



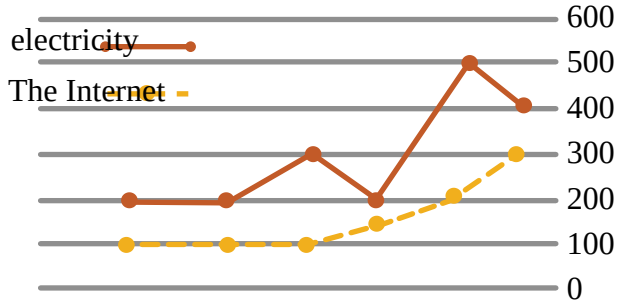
(A) 100 (B) 250 (C) 300 (D) 500

Answer: A

100 of the form is the most frequent value On the Internet it i

Legend training

What is the pattern of electricity bill values?



(A) 100 (B) 200 (C) 250 (D) 500

26 9? Numbers is 6 if the mode is l

62, S (among these numbers whose sum is 8.8) and it was

Then x = ...

A 6 B 7 C 10 D 19

Answer: D

Repeated twice: 8.9 = mode

Repeated more than twice9

Numbers 9,9,9,, Q8,8

62 = sum of numbers

= ·· -)9+9+9+8+8(S

·· = ·· - ·· =

Legend training

, S among 7 and 7 and 8 numbers is 5 if the mode is for

So, x = 38. These numbers add up

A 6 B 8 C 10 D 19

4 The mediator

A It is the value in the middle between the two I'm after arranging it.

If the B The number of annas is even :

$$= \text{the mediator} \frac{\text{The sum of the middle two numbers}}{2}$$

B The median is equal to the average if the numbers

Increase or increase I mow regularly.

27 Trees in a garden 6 taller

How many times is 18, 20, 24, 36, 24, 19?

These lengths?

A 17 B 22 C 23 D 24

Answer: B

To find the mediator, we arrange the money Annat

26 , 24 , 24 , 20 , 19 , 18

$$= \text{the mediator} \frac{24 + 20}{2} = 22$$

Legend training

24, 26, 17, 24, 19. Trees in a garden are the highest heights.

What is the average of these lengths? 12, 20, 15, 18

A 19 B 20 C 21 D 24

28 Arat is a 4 grade student in Hide 90, 85 , 85 , ··

Then the school deleted her lowest grade

Compare between:

First value: average

The second value A: The mediator

A The first value is greater B The second value It's bigger

C The two values ··are equal D Data are insufficient

Answer: A

The order of the B Annat Delete the smallest 90, 85, 85, 70:

Albi Annat 90 ,85 , 85

$$= \text{average} \frac{90 + 85 + 85}{3} = \frac{2}{3}86$$

median = 85

The first value is greater

Legend training

Arat is a 4 grade student in Hide ·· , ·· , ·· , ··

Then the school dropped her highest grade.

Compare between:

First value: average

The second value A: The mediator

A The first value is greater B The second value It's bigger

C The two values ··are equal D Data are insufficient

29 7 and the model 2 are five natural numbers, the smallest

What is their median?6 and the average

A 5 B 6 C 7 D 8

Answer: C

average = 6 = 5

30 = their sum = their average x their number

2 patterns that will be repeated at least twice, the smallest of which is 7

Numbers can be 7 , 7 , 7 , 7 , 2

∴ The mediator 7

Legend training

6 and the model 3 are five natural numbers, the smallest of which

What is their median? 8 And the average

(A) 5 (B) 6 (C) 7 (D) 8

30 5 and the average 2 are five numbers, the smallest of which

Find the missing number 6, the median 7 and the mode

(A) 5 (B) 4 (C) 3 (D) 2

Answer: C

average = 5 = 5

Their total = 5 x 5 = 25

Repeat at least twice the pattern 7

6 is the number in the middle

7, 7, (6), (), (S2) numbers

S = 25 · 2) + 6 + 7 + (7

= 25 - 22 = 3

Legend training

8, the mode 6, and the average 3 are five of the smallest numbers

Find the missing number 7 and the median

(A) 7 (B) 4 (C) 3 (D) 2

31 Family 20? What is the median number of children in?

(A) 6 (B) 5 (C) 4 (D) 3

1	7	2	2	3	4
2	4	1	5	3
5	3	3	5	2
3	4	3	1	6

Answer: D

To find the mediator, we arrange the money I notice their n

= the mediator $\frac{\text{The sum of the middle two numbers}}{2}$

We arrange them to the eleventh term

3 , 3 , 3 , 3 , 3 , 2 , 2 , 2 , 1 , 1 , 1

= the mediator $\frac{3+3}{2}$ = 3

Legend training

Family 20? What is the median number of children in?

(A) 6 (B) 5 (C) 4 (D) 3

4	1	3	3	4
2	4	4	5	3
5	3	4	5	2
3	4	4	5	6

3 Odds

1 Counting principle

Number of selection methods R =

The product of multiplying the number of ways for each choice separately

4 Example: Restaurant serving meals from Vegetable varieties

Dream varieties and salad varieties, how many 3 and

Different meals that can be chosenAre there any of them?

meals of number = 3 x 4 x 2 = 24 meal

32 Three friends went to the cinema in some ways

3 chairs in a row? They can sit on

A 6 B 7 C 9 D 12

Answer: A

6 = 1 x 2 x 3 = number of ways

Legend training

Four friends went to the cinema in many ways

4 chairs in a row? They can sit on

A 24 B 18 C 16 D 12

33 10 doors in how many ways can a worker in a factory

Entering and exiting through two different doors?

(Entering through one door and exiting through another)

A 10 B 90 C 100 D 200

Answer: B

10 = number of entry methods

Because the entrance door does not allow him 9 = the number of exit ways

By getting out of it

Number of ways = 10 x 9 = 90

Legend training

Doors: How many ways can a worker enter a factory?

And leaving through two different doors?

(Entering through one door and exiting through another)

A 10 B 56 C 64 D 80

34 10 doors in how many ways can a worker in a factory

Enter and exit through which door?

A 10 B 90 C 100 D 200

Answer: B

10 = number of entry methods

10 = number of exit ways

= 10 x 10 = 100 Number of ways

Legend training

Doors in how many ways can a worker8 in a factory

Enter and exit through which door?

A 10 B 56 C 64 D 80

35 In how many ways can a password be created?

(Without 9, 7, 4, 2, 1) cells using the numbers 3 from

Repeat?

A 45 B 50 C 60 D 65

Answer: C

Numbers and no repetition 5 available options

The number of ways of the first cell Numbers5=

Number of ways to enter the second Oh Numbers 4 =

3 = the number of ways to enter the third digit numbers

Number of password methods = 5 x 4 x 3 = 60

Legend training

In how many ways can you create a two-digit password?

(Without repetition? 9, 7, 4, 2, 1) using numbers

A 45 B 30 C 20 D 25

36 In how many ways can a password be created?

(9, 7, 4, 2, 1) are digits using the numbers 3 from

With repetition?

A 336 B 125 C 320 D 340

Answer: B

ways of number = 5 x 5 x 5 = 125

Legend training

In how many ways can you create a two-digit password?

(With repetition? 9,7,4,2,1) using numbers

A 45 B 30 C 20 D 25

37 I see 1 student in 3 schools Rows, I want my sister

Form a team from the top three students in each class

For outstanding students, in how many ways can a team be formed?

A 9 B 27 C 54 D 81

Answer: B

A My sister R who? Student for each class 3

My sister's ways I see a first grade student = 3

My sister's ways I am a second grade student = 3

My sister's ways I am a third grade student = 3

Number of ways = 3 × 3 × 3 = 27

Legend training

I see 1 student in 4 schools Rows, I want my sister

From the top three students in each class to form a team

For outstanding students, in how many ways can a team be formed?

A 9 B 27 C 64 D 81

38 What is the difference between the largest number and th

less than 1000 (so that they are greater than 4, 3, 2, 1) numbe

repetition ?

A 2387 B 3287 C 3178 D 3087

Answer: D

To find the largest number, arrange the numbers from largest

4321 from the left

1234 and the smallest number vice versa

the difference ····= ···· - ···· =

Legend training

What is the difference between the largest number and the sn

Without repetition? 1000 (so that they are greater than 5, 4, 2

A 4176 B 4288 C 3178 D 3087

39 How many two-digit numbers can be formed? E from

(If repetition is allowed? 9, 6, 5, 4, 2) numbers

A 5 B 10 C 20 D 25

Answer: D

5 = the number of ways in the first digit

Number of ways to enter the second Oh = ·

Number of ways = 5 x 5 = 25

Legend training

How many two-digit numbers can be formed? It's numbers

·,·,· ,·,·( ), if repetition is not allowed

A 20 B 22 C 23 D 24

2 Repent Adele

Used during sisterhood R, taking into account the order

-Words- (such as functions numbers)

Example:

$^{5}_{2}$ to Start with the highest number and write two numbers: 4 x 5 = Repent

$^{7}_{3}$ to It started with 5×6×7 = Numbers 3 and we write 7

40 My employees are 3 years old. How can I choose a director and a secretary?

The first shall be a director and the second shall be 8 Randomly

An agent and the third is a secret agent?

A 60 B 96 C 220 D 336

Answer: D

The choice specifies the question function Adele

$^{8}_{3}$ to = 8 x 7 x 6 = 336

Legend training

My employees are among 3. How can I choose a company manager?

The first shall be a director and the second shall be 10 Randomly

An agent and the third is a secret agent?

A 60 B 100 C 220 D 720

3 Combinations

It is used during sisterhood R set of set

Amplify without taking into account the order.

Example: $^{5}_{2}Q = \frac{4 \times 5}{1 \times 2} = 10$

Repent Add in the numerator and denominator the factorial of the number

Bottom $^{8}_{3}Q = \frac{6 \times 7 \times 8}{1 \times 2 \times 3}$ 56=

41 3 employees How many ways to choose a company director

To go to a job meeting? 5 Son Randomly

A 12 B 10 C 8 D 6

Answer: B

My sister I did not respect the order

The question is combinations

$^{5}_{3}Q = \frac{3 \times 4 \times 5}{1 \times 2 \times 3} = 10$

Legend training

How many 4 employees choose a company director?

To go to a business meeting? 6 Ben Randomly

A 22 B 20 C 15 D 6

4 = probability of the event $\frac{\text{Number of times the event occurr}}{\text{The number of elements of the eye sp}}$

42 When you roll a number cube, what is the probability of

4 Greater than?

A $\frac{1}{2}$ B $\frac{1}{6}$ C $\frac{1}{3}$ D $\frac{1}{4}$

Answer: C

Eye space Oh = { 6 , 5 , 4 , 3 , 2 , 1 }

= 4 Event = a number greater than { 6 , 5 }

Probability = $\frac{2}{6} = \frac{1}{3}$

Legend training

5 When throwing a number cube, what is the probability of a

A $\frac{1}{2}$ B $\frac{1}{6}$ C $\frac{1}{3}$ D $\frac{1}{4}$

43 10 to 1 A box containing cards numbered from

I pulled a card from him, so what is the probability of it being

3 Divide by?

A $\frac{1}{10}$ B $\frac{3}{10}$ C $\frac{7}{10}$ D $\frac{1}{2}$

Answer: B

Eye space Oh = { 10 ,.... 3 , 2 , 1}

= 3 is a number that accepts {9 , 6 , 3}

Probability = $\frac{3}{10}$

Legend training

I pulled from him 20 to 1 box containing cards numbered from

4 cards, what is the probability that it is a number divisible by?

A $\frac{1}{10}$ B $\frac{1}{4}$ C $\frac{7}{10}$ D $\frac{1}{2}$

44 A ball was drawn from a numbered box 20 to 1

What is the probability that the drawn number is an odd number?

A %20 B %30 C %40 D %50

Answer: D

Eye space Oh { 20 ,...., , 3 , 2 , 1}

= odd numbers 19 , ... , 7 , 5 , 3 , 1}

= probability 20 $\frac{10}{2}$ = 1 = 50 %

Legend training

What is the ratio of 20 to 1? A ball is drawn from a box numbered from

What is the probability that the drawn number is a prime number?

A %20 B %30 C %40 D %50

45 White, 5 yellow, 3 red balls, and 4 boxes in it

What is the probability of drawing a random non-white ball? A?

A $\frac{5}{12}$ B $\frac{7}{12}$ C $\frac{3}{12}$ D $\frac{4}{12}$

Answer: B

12 = number of balls

7 = non-white balls

= non-white probability $\frac{7}{12}$

Legend training

White balls, 5 yellow balls, 6 red balls, and 4 boxes in it

The probability of drawing a non-red ball is random A?

A $\frac{11}{15}$ B $\frac{12}{15}$ C $\frac{13}{15}$ D $\frac{4}{15}$

46 A red, yellow and green ball and the probability of dr

Random red ball A $\frac{2}{5}$ How many yellow balls

And green?

A 5 B 10 C 15 D 20

Answer: C

25 = number of balls

= red probability $\frac{2}{5}$

= Probability of hitting $\frac{3}{5}$ balls

Number of yellow and green balls = $\frac{3}{5}$ × 25 = 15 Ball

Legend training

A red, yellow and green ball and the probability of drawing

Random red ball A $\frac{2}{5}$ How many yellow balls are there?

And green?

A 8 B 12 C 18 D 20

47 A student from the first year of secondary school and

I am in the second year of secondary school in school radio

A student from the second year of secondary school?

A $\frac{1}{2}$ B $\frac{3}{5}$ C $\frac{1}{3}$ D $\frac{2}{5}$

Answer: D

The number of students is five

The Taliban2 students

Probability $= \frac{2}{5}$

Legend training

A student from class 3, a student from the first year of secondary school, and 5 were nominated

Secondary school radio station, what is the probability of my graduation? I'm a student

From the second year of secondary school?

(A) $\frac{1}{2}$ (B) $\frac{3}{8}$ (C) $\frac{1}{3}$ (D) $\frac{2}{5}$

4 Circular sectors

The first case:

The data is an angle to the figure

(1) When asked about the angle:

360 We add everyone and subtract them from

Luck:

(1) 90 = quarter circle

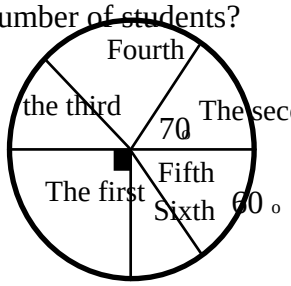
(2) semicircle = 180 °

(3) 270 = three-quarters of a circle

(4) Take into account the relationships between angles

48 The corresponding figure shows the number of students in a school

The class with the smallest number of students?



(A) The first (B) The second (C) Fourth (D) Sixth

Answer: D

The first is an angle = quarter circle ∵ = °

third = fifth = 60 Head to head

fourth = (-180 60 ° ∵ + °) = ∵ °

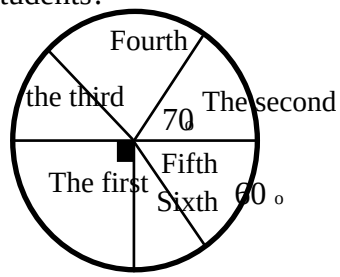
sixth = (-180 90 ° ∵ + °) = ∵ °

The lowest is sixth

Legend training

The corresponding figure shows the number of students in a s

Who has the largest number of students?



(A) The first (B) The second (C) Fourth (D) Sixth

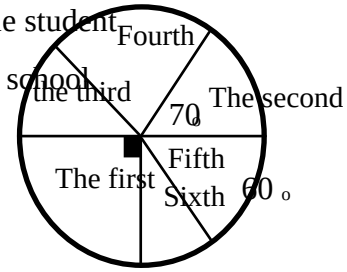
(2) Asking about the number

$= \text{number} \frac{\text{Corner}}{360} \times \text{Total total}$

49 If the number is a female student

There are 120 students in the school

Third grade student?



(A) 30 (B) 20 (C) 15 (D) 10

Answer: B

60 = corner of the third row = fifth °

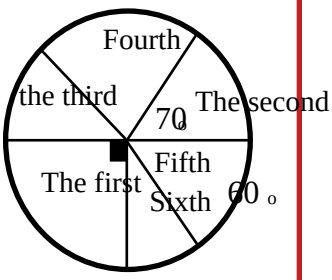
(head to head)

$= \text{Number of third grade students} \times \frac{60}{360} \; 120 = 20$

Legend training

If the number of students in the school is 180, what is the number of students?

Fourth grade?



(A) 30 (B) 25 (C) 15 (D) 10

3 Question about the percentage

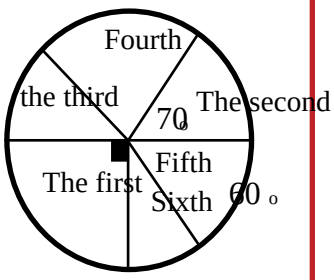
$$= \text{percentage} \quad \frac{\text{Corner}}{360} 100\times$$

50 What is the percentage?

Fourth grade student

!

And a fifth is rounded A ?



(A) %··,· (B) % 35 (C) %44.5 (D) %50

Answer: B

row fourth the of corner = 180 - 60) + (70

50 = 130- 180=

··· = ·· + ··Angle of the fourth and fifth

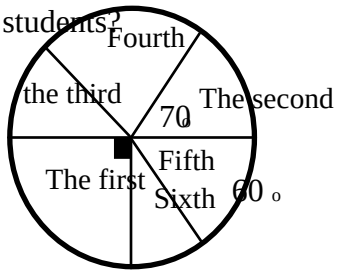
$$= \text{percentage} \quad \frac{110 \text{ x } 100}{360} = \frac{1100}{36} \quad \% 30$$

Legend training

What is the percentage of students?

Fourth and second grade

Rounding A?



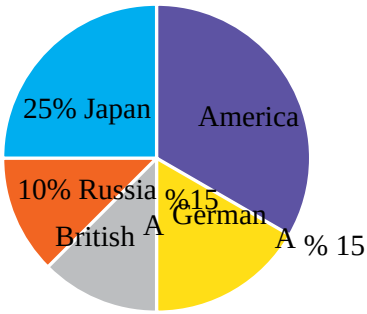
(A) %··,· (B) % 35 (C) %44.5 (D) %50

Second referral A: If a percentage is given in the figure

1 $$\text{Sector angle} = \frac{\text{Sector ratio}}{100} °360\times$$

5 In the corresponding figure, what angle does Scan repr

Russia ?



(A) 22 (B) 30 (C) 36 (D) 45

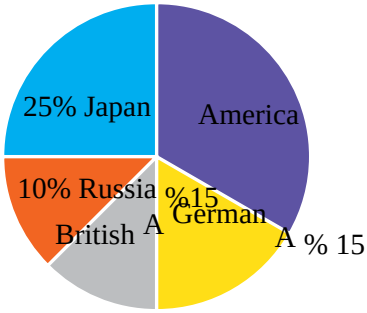
Answer: C

$$= \text{sector angle} \frac{10}{100} \times 360 = 36$$

Legend training

In the corresponding figure, what angle does it represent?

British population A?



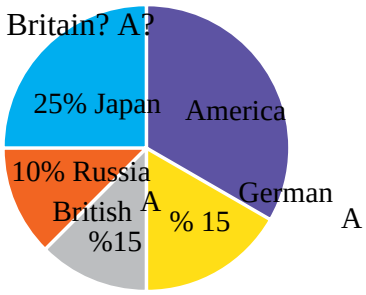
(A) 22 (B) 30 (C) 36 (D) 54

2 If the question is about the numerical value of the sector :

$$\text{numerical value} = \frac{\text{Sector ratio}}{100} \times \text{Total total}$$

52 1000000 in the corresponding fee if the population.

What is the population of Britain? A?



A 12000 B 140000 C 150000 D 400000

Answer: C

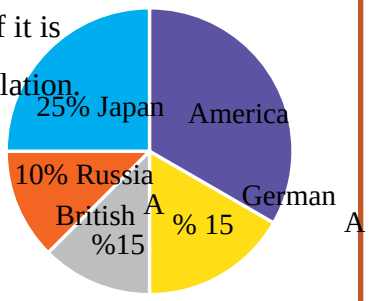
$$S = \frac{15}{100} \times 1000000 = 150000$$

Legend training

In the corresponding fee if it is

How many 2000000 population.

British population A?



A 12000 B 140000 C 150000 D 300000

3 If the question is about the percentage :

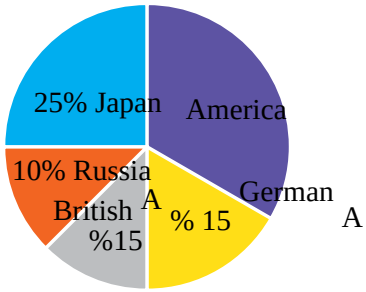
100% If you find all the ratios, add them and subtract them

Luck:

%50 = semicircle, %25 = quarter of the circle

75% = three-quarters of the circle

53 In the chart Shown is the percentage of America's po



A %45 B %35 C %25 D %15

Answer: B

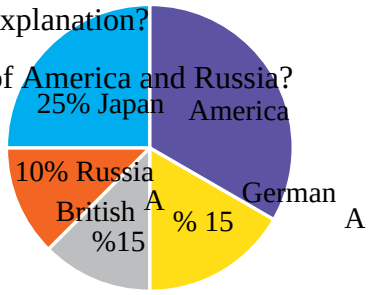
percentage America·s = 100 - 25) + 10 + 15 + (15

= 100 65- = %35

Legend training

In the diagram Where is the explanation?

What is the population ratio of America and Russia?

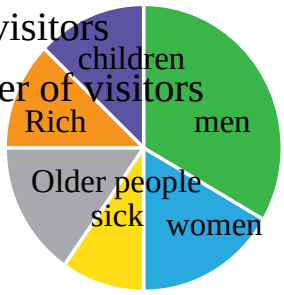


A %45 B %35 C %25 D %15

Third case: If you give the number .

1 Sector angle $= \frac{\text{Numerical sector value}}{\text{Total total}} \times 360°$

54 If the number of visitors 720 people and number of visitors Find 140 children Angle E?



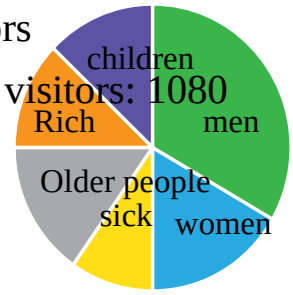
A 40 B 50 C 60 D 70

Answer: D

= sector angle $\frac{360 \times 140}{720}$ = 70 °

Legend training

If the number of visitors People and number of visitors: 1080 Find 120 children Angle E?

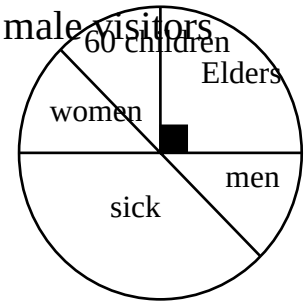


A 35 B 40 C 45 D 70

2 If the question is about the number

They are added and subtracted from the whole number

55 If the number of visitors is 560 people in the corresponding figure Find the number of male visitors



A 80 B 90 C 100 D 120

Answer: A

= Sheikhs Circle1 = $\frac{1}{4}$ x 560 = 140

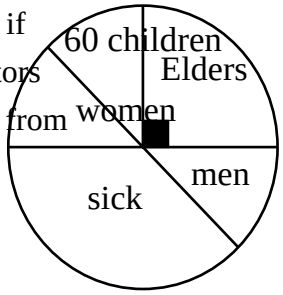
= Children+elders+men $\frac{1}{2}$ Circle

men + 140 + 60 = 280

men = 280 - 200 = 80

Legend training

In the corresponding figure if 560 people, number of visitors Find the number of visitors from Sick?

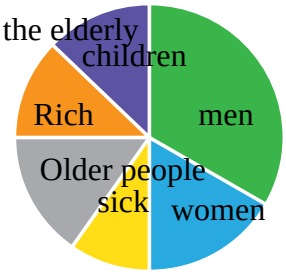


A 60 B 90 C 180 D 220

3 Question about the percentage

= percentage $\frac{\text{Sector value}}{\text{Total total}}$ 100×

56 The number of visitors is 80 if the number of visitors is Find their percentage? 12 of the elderly



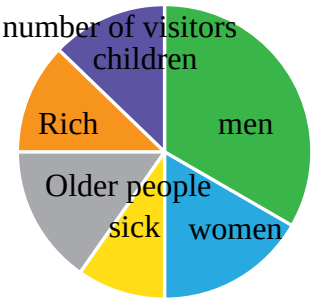
A %12 B %15 C %17 D %20

Answer: B

= percentage $\frac{12}{80}$ 100× = 15%

Legend training

And the number is 80 if the number of visitors 16 elderly visitors Find their ratio?

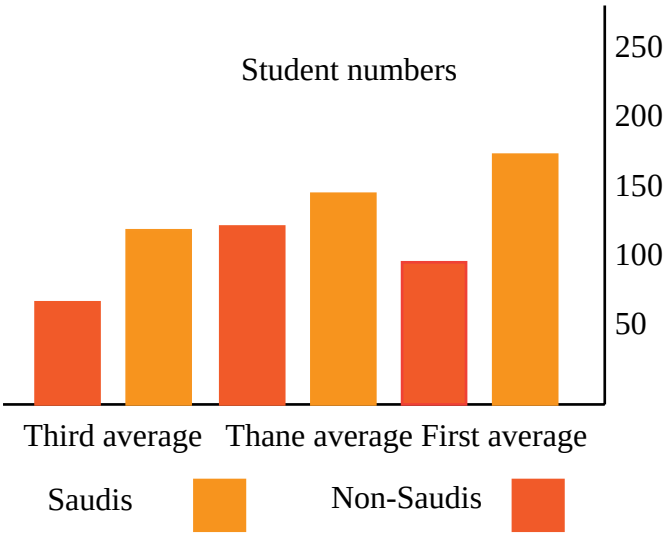


A %12 B %15 C %17 D %20

5 Reading graphics

57 Representation of Alpi This shows the numbers of students in Uhud Schools, what is the percentage of third grade middle school students?

Almost the entire school A ?



(A) %20 (B) %25 (C) %30 (D) %50

Answer: B

= Number of third grade students ⋯ + 125 = ⋯

= Number of students in the school ⋯ + 275 + ⋯ = ⋯

= ratio $\frac{200}{750} \times 100 = \frac{80}{3}$ 26

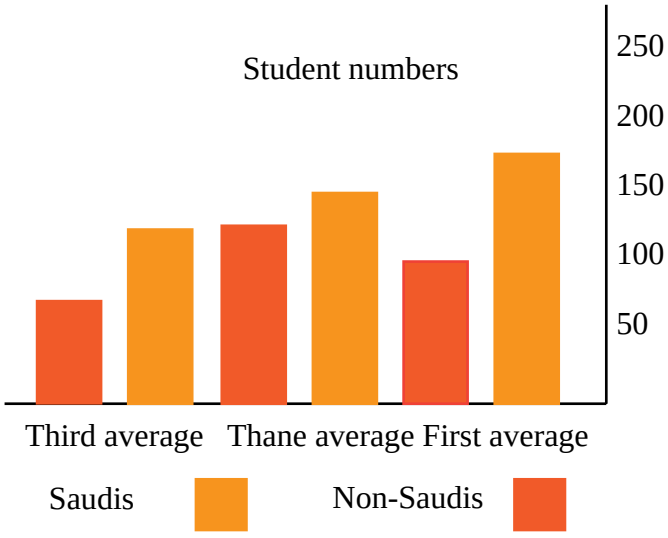
25 = closest answer

Legend training

Representation of Alpi This shows the number of students i

What is the ratio of the second intermediate grade to the ent

Approximately a ?



(A) %20 (B) %25 (C) %35 (D) %50

58 Which sports are the most popular? In the four cities?

	Riyadh	grandmother	Dammam	Abha
3524	Basketball	....	....	....
3451	Football	....	....	....
2776	hand punch	....	....	....
1652	Volleyball	....	....	....
758	Water polo	...	...	...

(A) basketball (B) soccer

(C) Handball (D) water polo

Answer: A

Looking at the table we notice a basketball class at each B H

Higher than other sports

Legend training

Which sports are the least popular? In the four cities?

	Riyadh	grandmother	Dammam	Abha
3524	Basketball	....	....	....
3451	Football	....	....	....
2776	Handball	....	....	....
1652	Volleyball	....	....	....
758	Water polo	...	...	...

(A) basketball (B) soccer

(C) Handball (D) water polo

59 Use the table to answer the following questions Oh :

Sector	2016	General 2013	2014	2015
4132	Education	...	....	....
5123	Industry	1000	...	3223
2000	Health	....	....	....
2000	study	...	....	7000

2014 What is the rate of decline for the Bani Aim industry? 2015 and?

(A) %40 (B) %30 (C) %20 (D) %10

Answer: D

We go to industry class

1000 was the result in 2014

900 was the result in 2015

Decrease ... = ... - .... =

= percentage of decrease 100× C your

= $\frac{100}{1000}$ x 100 = %10

Legend training

Use the table to answer the following questions A:

Sector	2016	General 2013	2014	2015
4132	Education	...	....	....
5123	Industry	1000	...	3223
2000	Health	....	....	....
2000	study	...	....	7000

2016 and 2013 What is the rate of increase in health care ar

(A) %400 (B) %300 (C) %200 (D) %100

60 Use the table to answer the following questions Oh :

Sector	2016	General 2013	2014	2015
4132	Education	...	....	....
5123	Industry	1000	...	3223
2000	Health	....	....	....
2000	study	...	....	7000

What sectors declined?

(A) Education and industry (B) Health and industr

(C) Health and agriculture (D) Agriculture and edu

Answer: A

From the table we read the options, we find that education a

They are the sectors that declined Ajaha.

Legend training

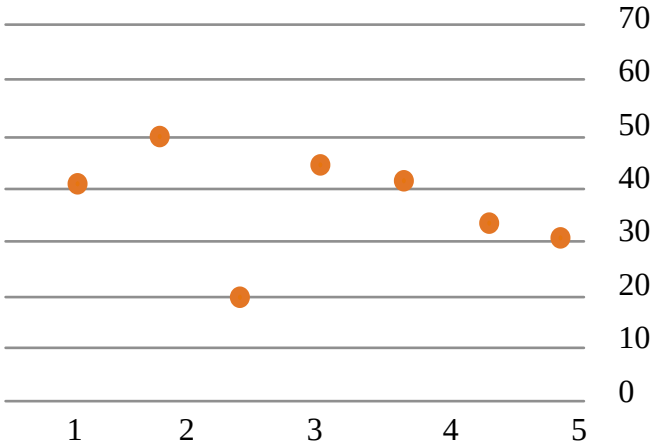
Use the table to answer the following questions A:

Sector2016	General		
	2013	2014	2015
4132Education	···	····	····
5123Industry	1000	···	3223
2000 Health	····	····	····
2000 study	···	····	7000

2014 What sectors increased in 2014?

A education B health
C agriculture D industry

61 Find the difference between the third and sixth day



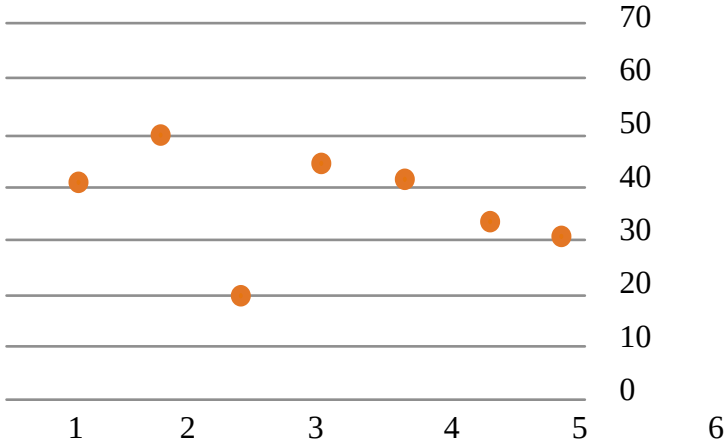
A 3 B 4 C 6 D 8

Answer: D

difference the = 50 - 42 = 8

Legend training

Find the difference between the first and fifth day



A 5 B 6 C 7 D 10

When reading the pi There are equivalen
A number or more = at least = a number
A number or less = at most = a number a
We only take the Largest number = Grea
Less than a number = we only take the le

62 From the following table, calculate the percentage of th
At least? 170

150 length	160	170	170.5	180	199
5number	5	10	5	15	10

A $\frac{4}{5}$ B $\frac{2}{5}$ C $\frac{1}{4}$ D $\frac{3}{7}$

Answer: A

And shorter than 170, at least you can designate those whose

··= + · + ·· + ·· = ··

= We find the ratio $\frac{40}{50} = \frac{4}{5}$

Legend training

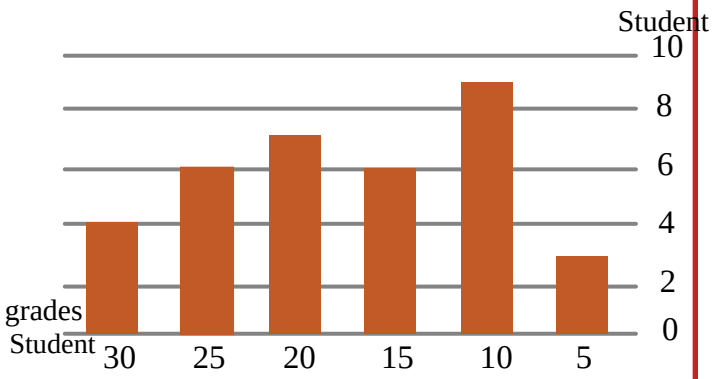
180 From the following table, calculate the percentage of those who are tall

At least?

150 length	160	170	170.5	180	199
5number	5	10	5	15	10

(A) $\frac{4}{5}$ (B) $\frac{2}{5}$ (C) $\frac{1}{4}$ (D) $\frac{1}{2}$

(63) Grades or less10 The number of students who obtained



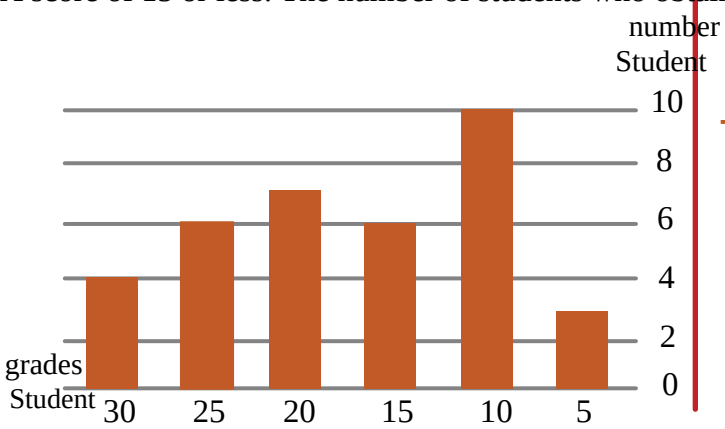
(A) 9 (B) 10 (C) 12 (D) 14

Answer: C

less or degrees 10 = 9 + 3 = 12

Legend training

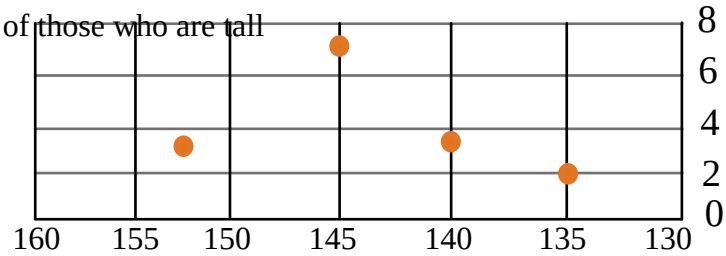
A score of 15 or less. The number of students who obtained



(A) 19 (B) 20 (C) 32 (D) 34

(64) From the drawing, find the number of people whose he

140?



(A) 9 (B) 12 (C) 10 (D) 14

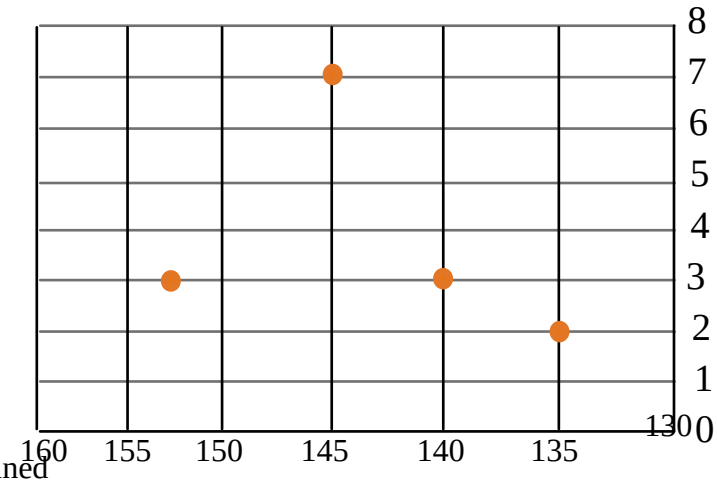
Answer: C

number 140 He will be counted from their height. 140 Their height

Analyze = · + · = ··

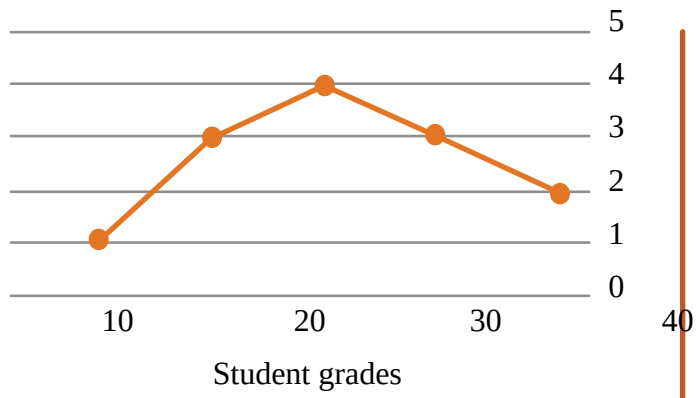
Legend training

From the drawing, find the number of people whose height



(A) 4 (B) 3 (C) 2 (D) 1

65 The corresponding figure shows the grades of a student in a class
The score of how many successful students is 40? If the success is from



A 13 B 5 C 4 D 3

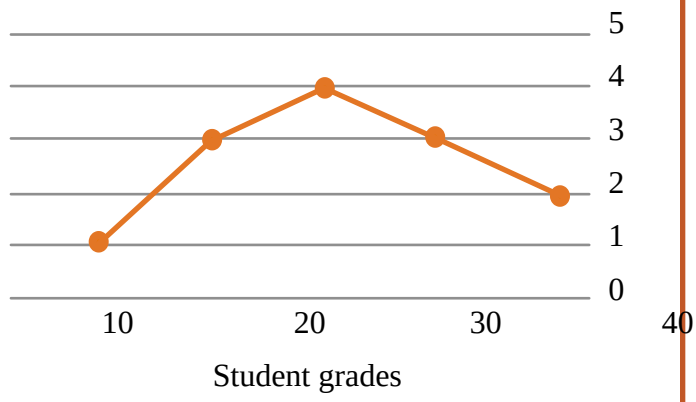
Answer: C

The number of successful students is 40 or more

analyze = 3 + 1 = 4

Legend training

The corresponding figure shows the grades of a student in a class. If...
How many students failed? Grade 40. Success from



A 13 B 12 C 9 D 5

66 How many 15 in the corresponding table does each sq
Fifth grade student?

the line	Abdul squares
Fourth	■ ■ ■
Fifth	■ ■ ■ ■ ■
Sixth	■ ■ ■ ■ ■ ■ ■

A 45 B 75 C 95 D 105

Answer: B

15 x the number of fifth grade students = the number of squ

= 5 x 15 = 75

Legend training

In the corresponding table, how many students does each sq

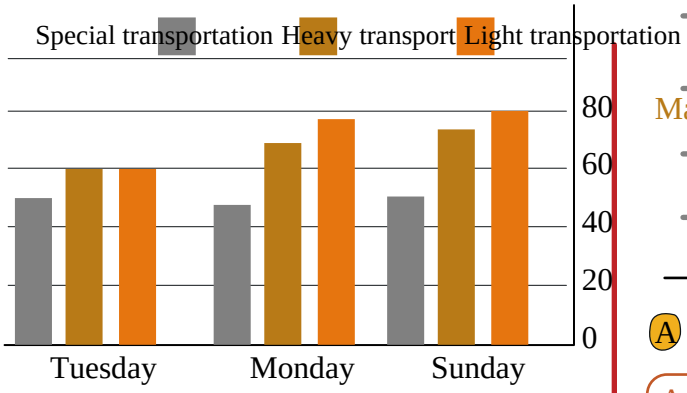
Sixth grade?

the line	Abdul squares
Fourth	■ ■ ■
Fifth	■ ■ ■ ■ ■
Sixth	■ ■ ■ ■ ■ ■ ■

A 45 B 75 C 95 D 105

67 If we want to avoid crowding in private transportation

And heavy, what day do we move?



A Sunday B Monday C Tuesday D Wednesday

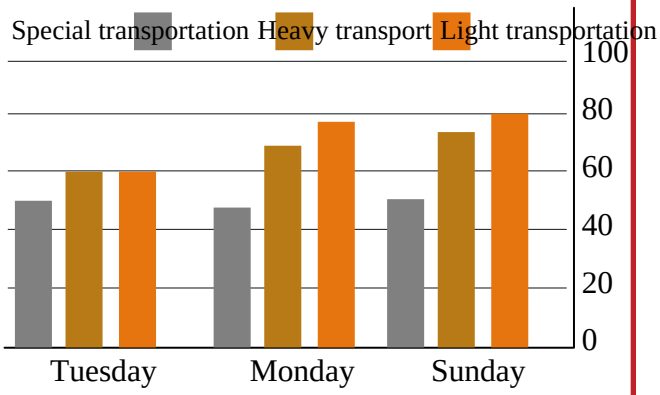
Answer: C

According The figure shows the lowest day in the number of private cars

The heavy one is Tuesday.

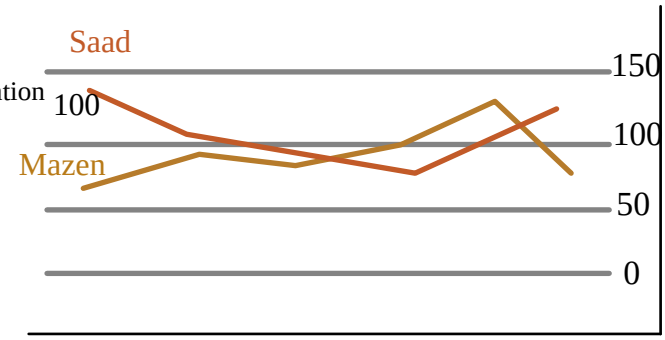
Legend training

What is the busiest day for light and heavy transport?



A Sunday B Monday C Tuesday D Wednesday

68 How many times does Majed's weight equal to Saad?



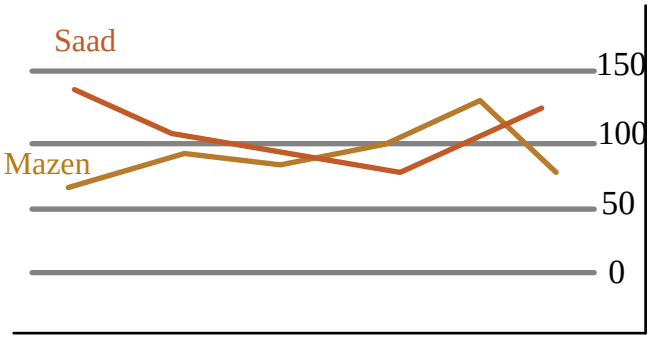
A time B She passed me C third D They are not e

Answer: B

From the drawing, the two shapes intersect twice

Legend training

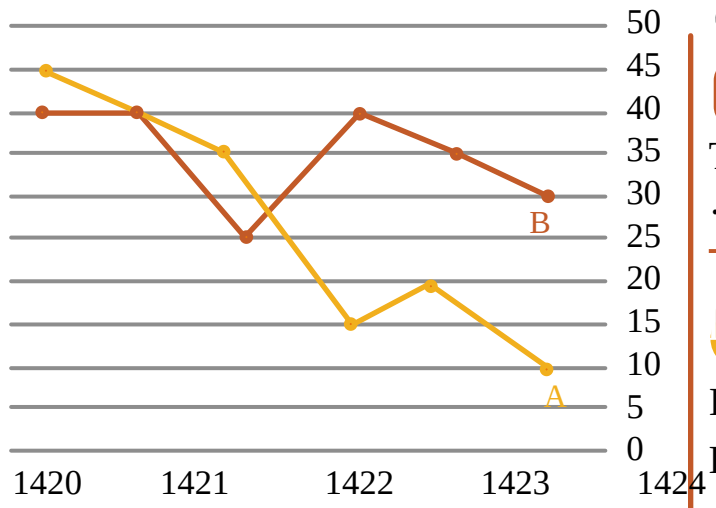
How much weight is Mazen equal to Saad?



A 100 and 70 B 150 and 80 C 110 and 90 D 140 an

69 The figure shows the price of two commodities. What is the difference in price between the two?

The highest value and lowest value of good B?



A SAR 10 B SAR 15 C SAR 20 D SAR 25

Answer: B

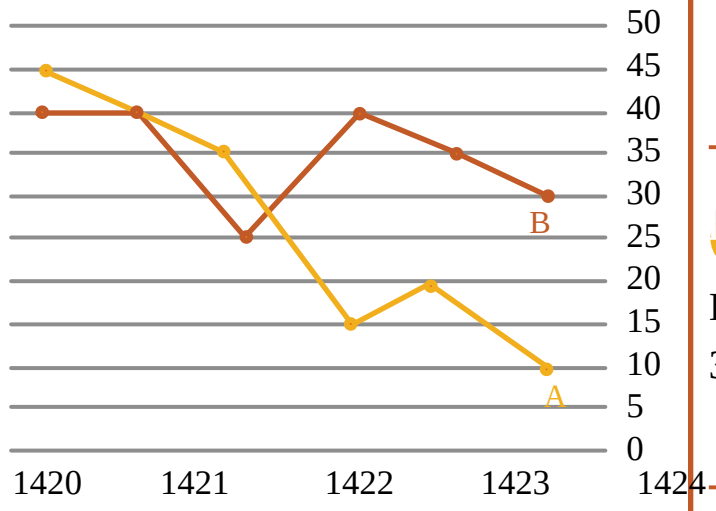
The difference between the highest value and the lowest value of commodity B

..=..-..

Legend training

The figure shows the price of two commodities. What is the difference in price between the two commodities?

What is the value and minimum value of commodity A?



A SAR 10 B SAR 15 C SAR 20 D SAR 35

70 In the previous figure, in which year was the

The price between the two goods?

A 1420 B 1421 C 1422 D 1424

Answer: B

The largest difference between the two commodi

.. = .. - ..

Legend training

In the previous figure, in which year was the sma

Between the two commodities?

A 1420 B 1421 C 1422 D 1424

71 In the previous figure, in what year was the

The price of good B to the price of good B as hig

A 1420 B 1421 C 1422 D 1424

Answer: A

The ratio between the two commodities

years 1420 $= \frac{30}{10} = 3$

years 1421 $= \frac{35}{20} = \frac{7}{4} = \frac{3}{4}$ 1

years 1422 $= \frac{40}{50} = \frac{8}{3} = \frac{2}{3}$ 2

The largest year ....

Legend training

In the previous figure, in what year was the ratio

3:1 Good A to the price of Good B equals

A 1420 B 1421 C 1422 D 1423

72 The table shows the number of students who graduated in each to divide.

Find the number of science students, second floor?-1

Specialization sciences E				Physics		chemistry		total	
The role 1	2	1	2	1	2	1	2	1	2
Student 35?	17		18	12	15	16	14	80	90

A 53 B 50 C 47 D 43

Answer: D

Number of science students, second round. We collect all the materials one round

Then we subtract them from the total for the second round

(-90= =)··+··+·· 90 47- = 43

Legend training

The table shows the number of students who graduated in each department

Find the number of science students?

Specialization sciences E				Physics		chemistry		total	
The role 1	2	1	2	1	2	1	2	1	2
Student 35?	17		18	12	15	16	14	80	90

A 53 B 50 C 68 D 78

73 In the previous table, what is the percentage of students?

Physics first round to the total number of first round students?

A %15 B %20 C %35 D %40

Answer: A

= ratio $\frac{80}{12} \times 100 = \frac{120}{8}$ = %15

Legend training

In the previous table, what is the percentage of chemistry stu

To the second-year student? Rounding A

A %15 B %20 C %35 D %40

74 In the previous table, the percentage among chemistry s

To all students

A $\frac{5}{18}$ B $\frac{3}{8}$ C $\frac{3}{7}$ D $\frac{3}{17}$

Answer: D

student chemistry of sum = 16 + 14 = 30

students all of sum = 80 + 90 = 170

= ratio $\frac{30}{170} = \frac{3}{17}$

Legend training

In the previous table, the percentage between students E deer

All student?

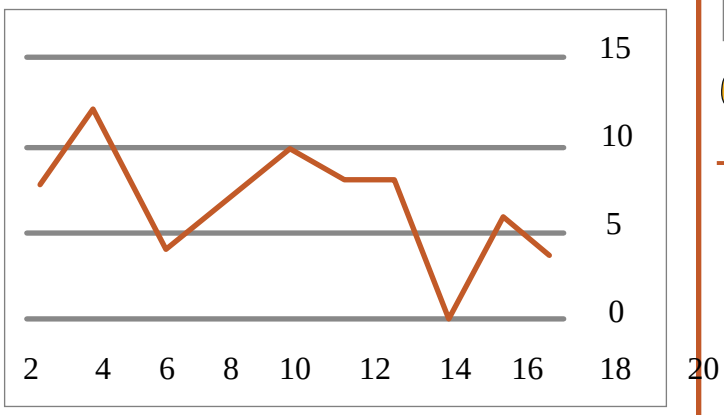
A $\frac{7}{34}$ B $\frac{3}{34}$ C $\frac{7}{17}$ D $\frac{3}{17}$

6 Read the drawing represen

When reading the pi I am confused

1. The line goes up when moving from the left For the right, there is an increase in the rate
2. Mix horizontally from left to right The rate is fixed
3. The line is down when moving from Left to right there are ten Cut in rate

75 For how many hours was the rate constant?



A 2 B 3 C 4 D 5

Answer: A

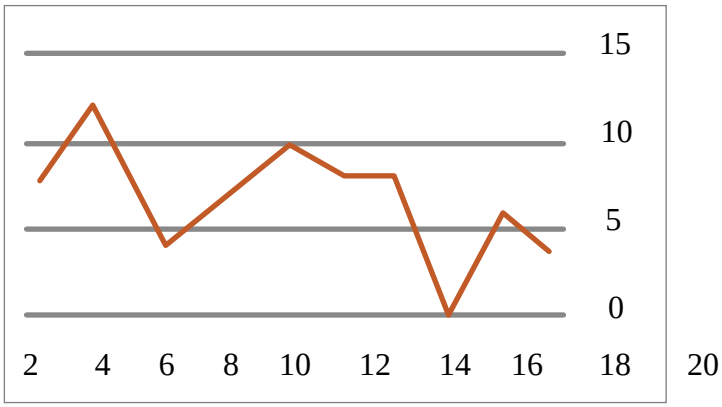
The rate is constant, which means that the line is horizontal

10 to 8 o'clock

The rate is constant over two hours

Legend training

For how many hours was the rate rising?



A 2 B 4 C 8 D 10